ReactJS

Introduction To ReactJS:

What is ReactJS?

ReactJS is a JavaScript library developed by Meta used for building fast and interactive user interfaces using reusable components.

Why Developers Like React?

1. Reusable code (as it is component based)
2. Faster websites (because of Virtual DOM)
3. Easy to Manage Large Projects

React vs Normal JavaScript

Normal JavaScript	React
More manual DOM work	Easier UI updates
Code becomes messy	Component-based
Hard to reuse	Reusable components
Slower for large apps	Better performance

Creating our first react app using create-react-app:

npx create-react-app my-app:

```
PS D:\ReactJS-revisited> npx create-react-app my-app

Creating a new React app in D:\ReactJS-revisited\my-app.
```

What is npx?

npx is a tool that comes with Node.js.

It allows you to:

- Run packages directly
- Without installing them globally

What is create-react-app?

create-react-app is a tool made to quickly create a React project with all setup already done.

Without it, you would manually configure:

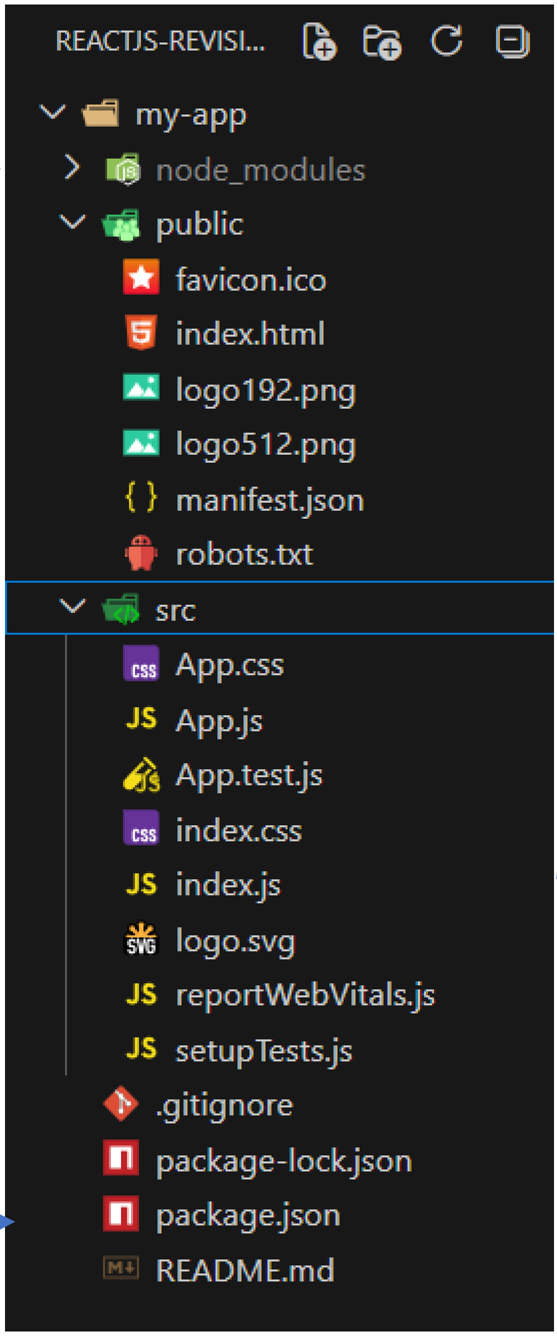
- Webpack
- Babel
- Dependencies
- Folder structure

This tool does everything automatically.

<u>What is my-app?</u>

This is your project name.

<u>What we get at the end in the folder?</u>



Node_modules: Contains all installed packages.

Important:

App.js: Main React component. Default starter code exists here.

index.js: Entry point of React app. This file connects React to the browser.

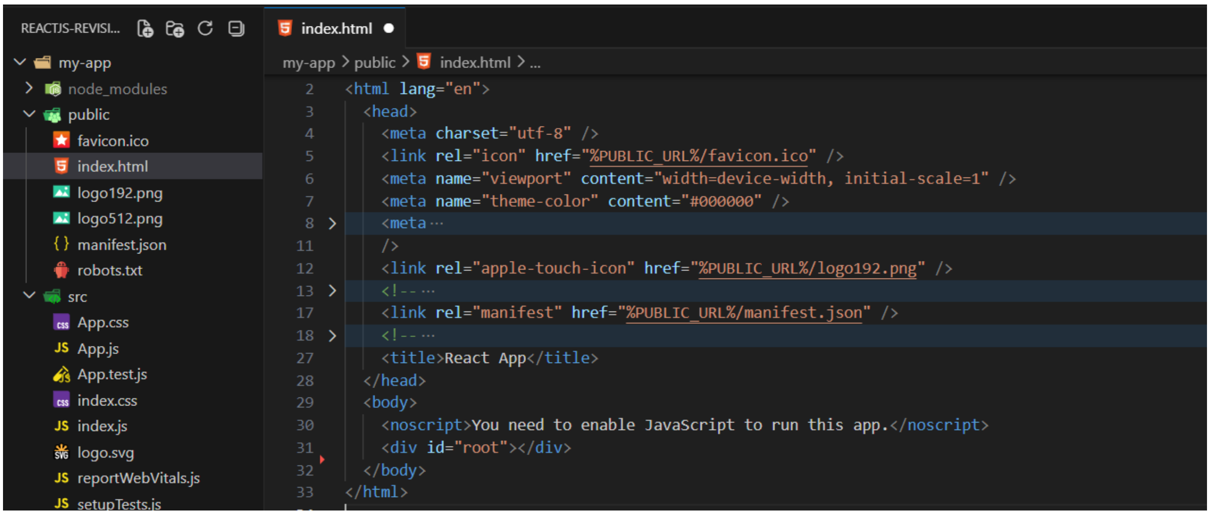
package.json: It Contains Project details , Installed packages , Scripts

Only “public” (Contains static files) and “src” (Main working folder) folder to be modified by us.

Two important files of “src” folder: App.js and index.js.

Exploring the ‘public’ folder:

Here the most important file is the “index.html”, rest can be ignored. We don’t modify the index.html even.

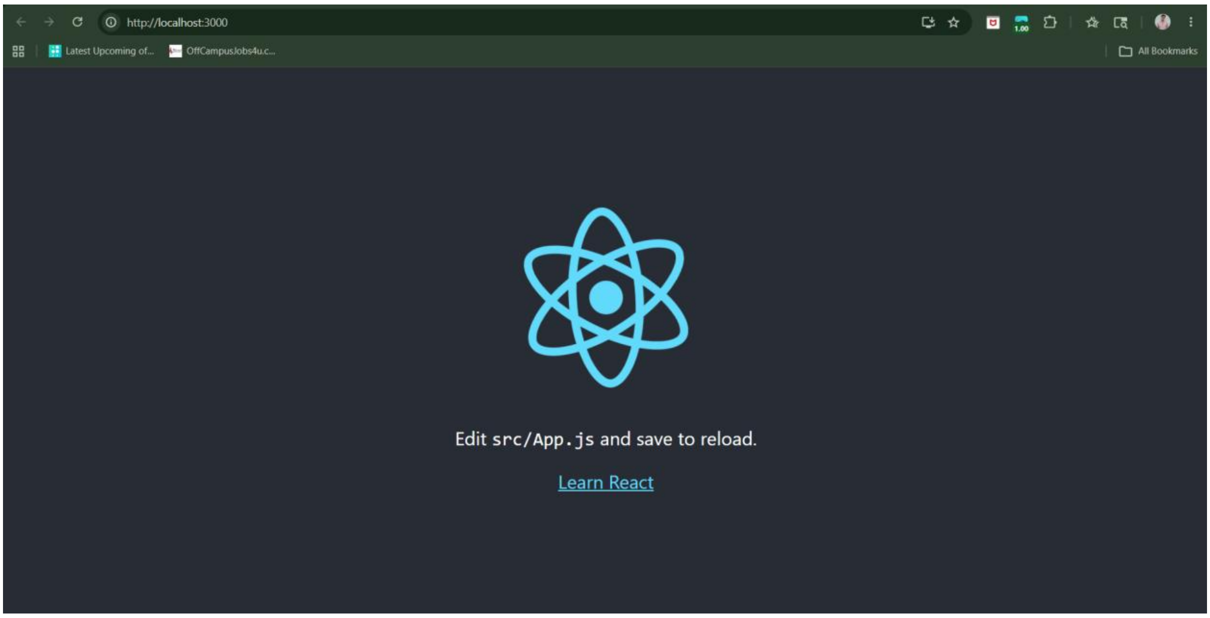


How to run the react app?

```
PS D:\ReactJS-revisited\my-app> npm start

> my-app@0.1.0 start
> react-scripts start
```

Output:



Understanding JSX

What is JSX?

JSX is a syntax used in React to write UI code in an HTML-like way inside JavaScript. It allows us to write:

- HTML-like code
- Inside JavaScript

Example:

const element = <h1>Hello React</h1>;

Is JSX same as HTML?

JSX looks like HTML, but:

- It is actually JavaScript
- Browser does not understand JSX directly
- React converts JSX into JavaScript

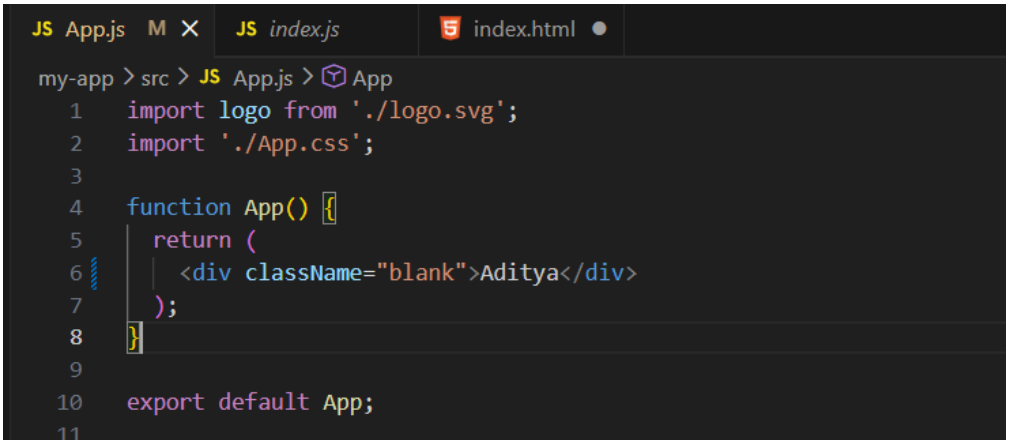
This conversion is done by:

- Babel (behind the scenes)

Why JSX is Useful?

1. Easier to read
2. Easier to write UI
3. JS + HTML together

Example: a basic example of JSX.



Here the App() is component, while <div> is JSX and, return() returns UI.

JSX rules:

Rule1: Return Only One Parent Element. Below example is correct as the <div> is the only one which is returned (discussed later in details)

```jsx
import logo from './logo.svg';
import './App.css';

function App() {
  return (
    <div>
    <h1>Utsav</h1>
    <div className="blank">Aditya</div>
    </div>
  );
}

export default App;
```

Rule 2: Every Tag Must Be Closed

```jsx
import logo from './logo.svg';
import './App.css';

function App() {
  return (
    <img/>
    // <br>
    );
}

export default App;
```

Here the <img> tag is closed, while the
 tag is not closed. If uncommented, it will show error.

Rule 3: Use className Instead of class

```jsx
import logo from './logo.svg';
import './App.css';

function App() {
  return (
    <div className="okay">Aditya</div>
    );
}

export default App;
```

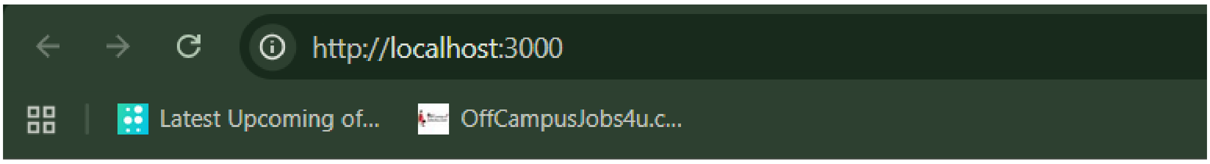
Rule 4: JavaScript Inside {} JSX allows JavaScript using curly braces.

```
JS App.js M ×    JS index.js    index.html ●
my-app > src > JS App.js > App > name
import logo from './logo.svg';
import './App.css';

function App() {
  const name = "Utsav";
  return (
    <div className="okay">{name}</div>
  );
}

export default App;
```

Output:



Utsav

<u>What is a Fragment?</u>

A Fragment is used to: Group multiple JSX elements without adding an extra HTML tag to the browser.

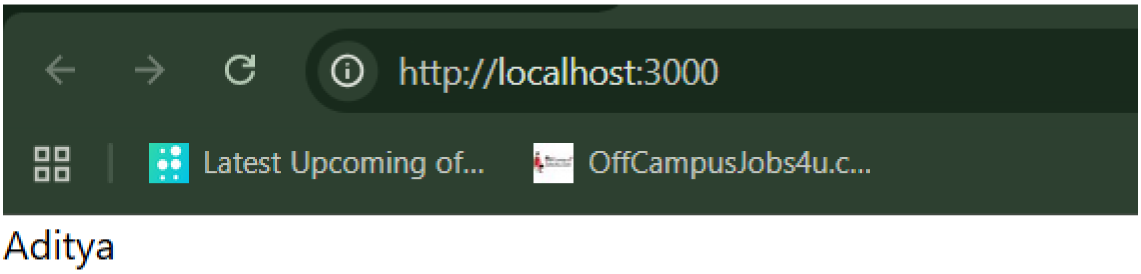
Without the <> </> (React Fragments):

```
JS App.js M ×    JS index.js    index.html ●
my-app > src > JS App.js > ...
import logo from './logo.svg';
import './App.css';

function App() {
  return (
    <div className="blank">Aditya</div>
  );
}

export default App;
```

Output:



In JSX, we must return: **Only one parent element.**

So, this is right:

```
JS App.js M ×   JS index.js   index.html ●
my-app > src > JS App.js > ...
import logo from './logo.svg';
import './App.css';

function App() {
  return (
    <div className="blank">Aditya</div>
  );
}

export default App;
```

While this is wrong: Error happens because React expects one root element.

```
JS App.js 1, M ×   JS index.js   index.html ●
my-app > src > JS App.js > default
import logo from './logo.svg';
import './App.css';

function App() {
  return (
    <h1>Utsav</h1>
    <div className="blank">Aditya</div>
  );
}

export default App;
```

<u>**How to deal with the above error?**</u>

Either you can make sure that only one root element is returned (by using one parent <div> or use the Fragments.

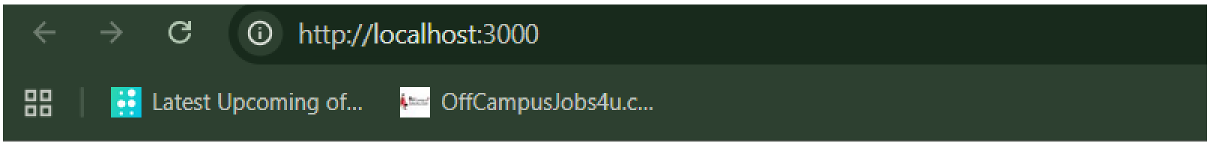
Example: using the parent <div>

```
JS App.js M ×    JS index.js    index.html ●
my-app > src > JS App.js > App
import logo from './logo.svg';
import './App.css';

function App() {
  return (
    <div>
    <h1>Utsav</h1>
    <div className="blank">Aditya</div>
    </div>
  );
}

export default App;
```

Output:



Utsav

Aditya

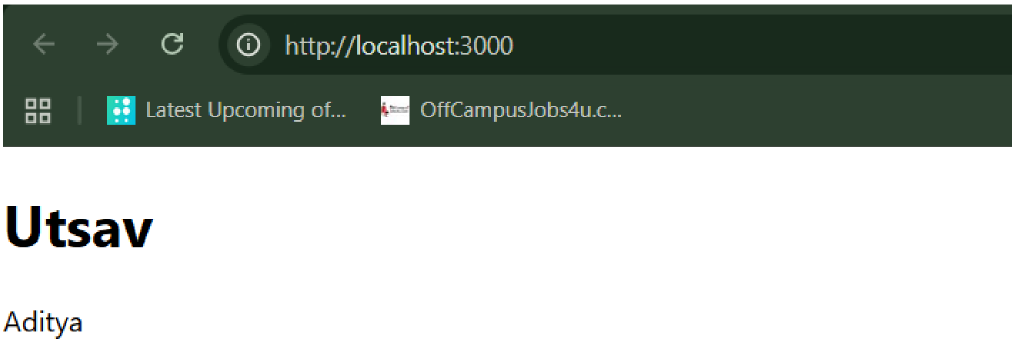
Example: using the fragments.

```
JS App.js M ×    JS index.js    index.html ●
my-app > src > JS App.js > ...
import logo from './logo.svg';
import './App.css';

function App() {
  return (
    <>
    <h1>Utsav</h1>
    <div className="blank">Aditya</div>
    </>
  );
}

export default App;
```

Output:



Setup + Adding Bootstrap to React:

Let's first understand about the types of page applications. By this we meant about the: Single Page Application (SPA) and Multi page application (MPA).

What is SPA?

A Single Page Application means: The website has only ONE HTML page, and content changes dynamically without full page reload.

Example: Gmail, Netflix, Instagram.

How SPA works?

When you open a SPA:

1. Browser loads one HTML file
2. React loads JavaScript
3. React controls the entire UI
4. When you click something:
    - Only part of the page updates
    - Page does NOT reload

Why SPA is Fast?

Because:

- Only required data changes
- No full page refresh
- Less network usage

Important File in SPA: public/index.html (This is the **ONLY HTML file** in React SPA. Everything else is JavaScript.)

What is MPA?

A Multi Page Application means: Every page is a separate HTML file, and browser reloads the page every time.

Examples: traditional websites, blogs, government websites.

How MPA Works?

When you click a link:

Home.html → reload → About.html → reload → Contact.html

Each page:

- Sends new request to server
- Loads full HTML again

Why MPA is Slower?

Because:

- Full page reload every time
- More server requests
- More loading time

SPA vs MPA:

Feature	**SPA**	**MPA**
Pages	One HTML page	Multiple HTML pages
Reload	No reload	Full reload
Speed	Fast	Slower
User experience	Smooth	Less smooth
Technology	React, Angular	Traditional HTML/PHP
SEO	Needs setup	Easier

Now, let's talk about the Node Modules folder:

What is node_modules folder?

node_modules is a folder that contains all installed packages (libraries) used in your project.

What is inside node_modules?

It contains:

- React library
- React DOM
- Babel tools

- Webpack tools
- Other dependencies
- Sub-dependencies (dependencies of dependencies)

Why node_modules folder is so big?

Because:

- Each library depends on other libraries
- All dependencies are installed locally

That’s why it becomes very large.

Why we don’t upload node_modules to GitHub?

Because:

- Very large (MBs to GBs)
- Can be recreated using package.json

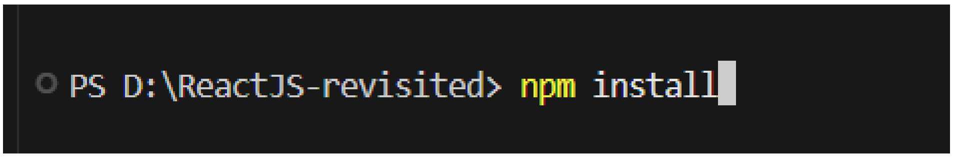
Can we delete the node modules folder?

Yes, we can.

How to install node_modules folder again if got deleted?

Using the following command:

npm install



Now, let’s talk about BootStrap in React.

What is Bootstrap?

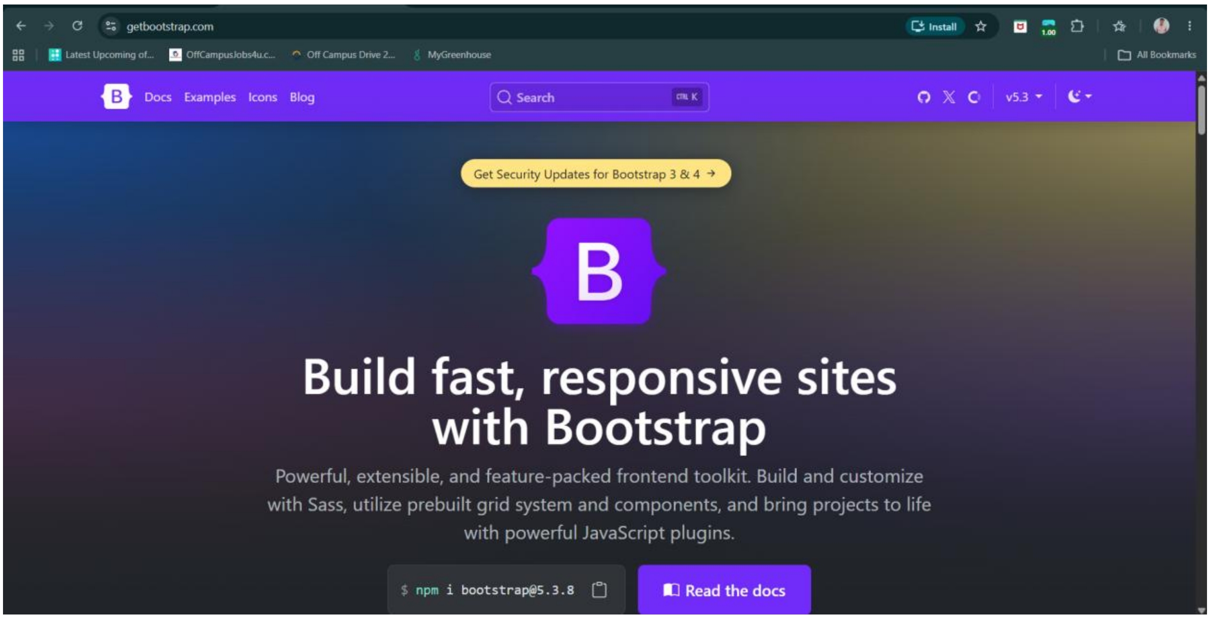
Bootstrap is a CSS framework that gives you:

- Ready-made styles
- Buttons
- Forms
- Navbar
- Grid system

Instead of writing CSS from scratch, you use prebuilt classes.

Where can we find this?

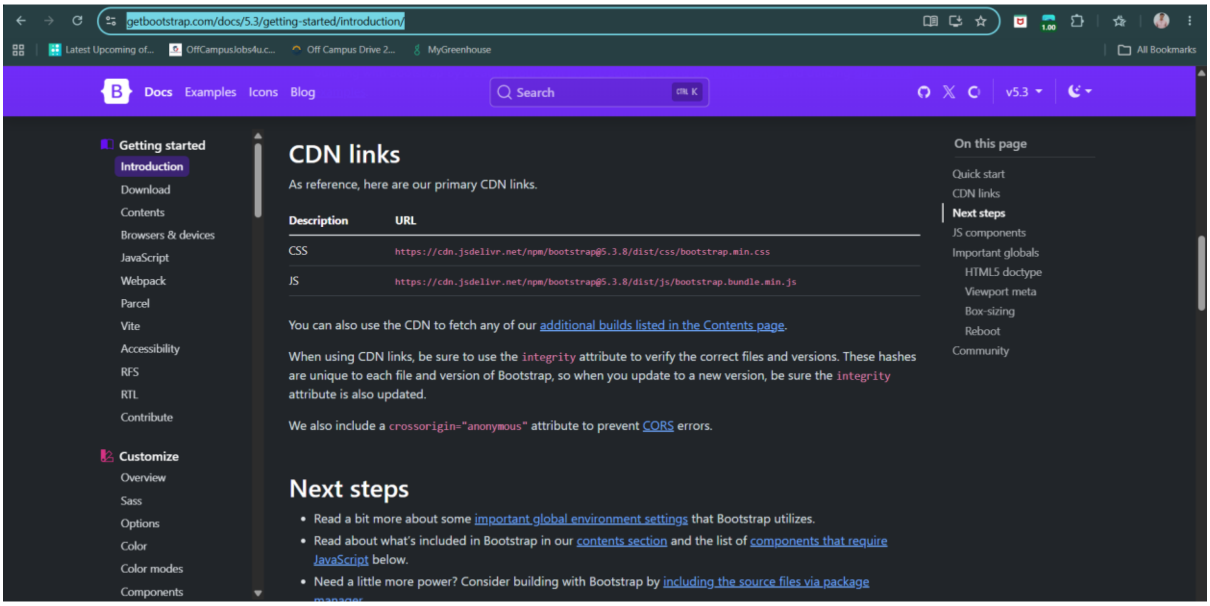
Website: https://getbootstrap.com/



How to include this BootStrap in our project?

Visit the docs:

https://getbootstrap.com/docs/5.3/getting-started/introduction/



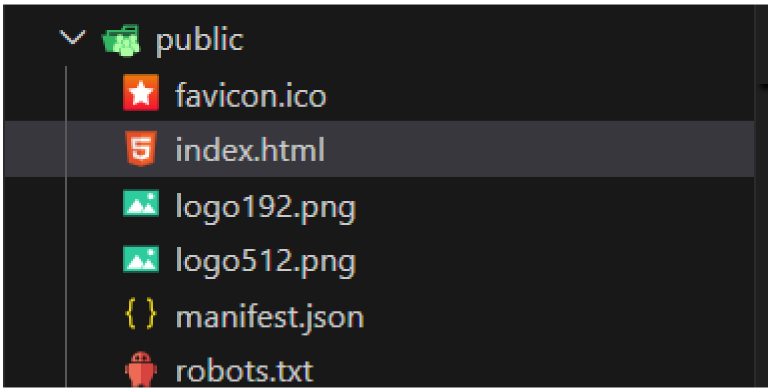
Using the CDN link:

`https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/js/bootstrap.bundle.min.js`

https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/css/bootstrap.min.css

Where to paste this link?

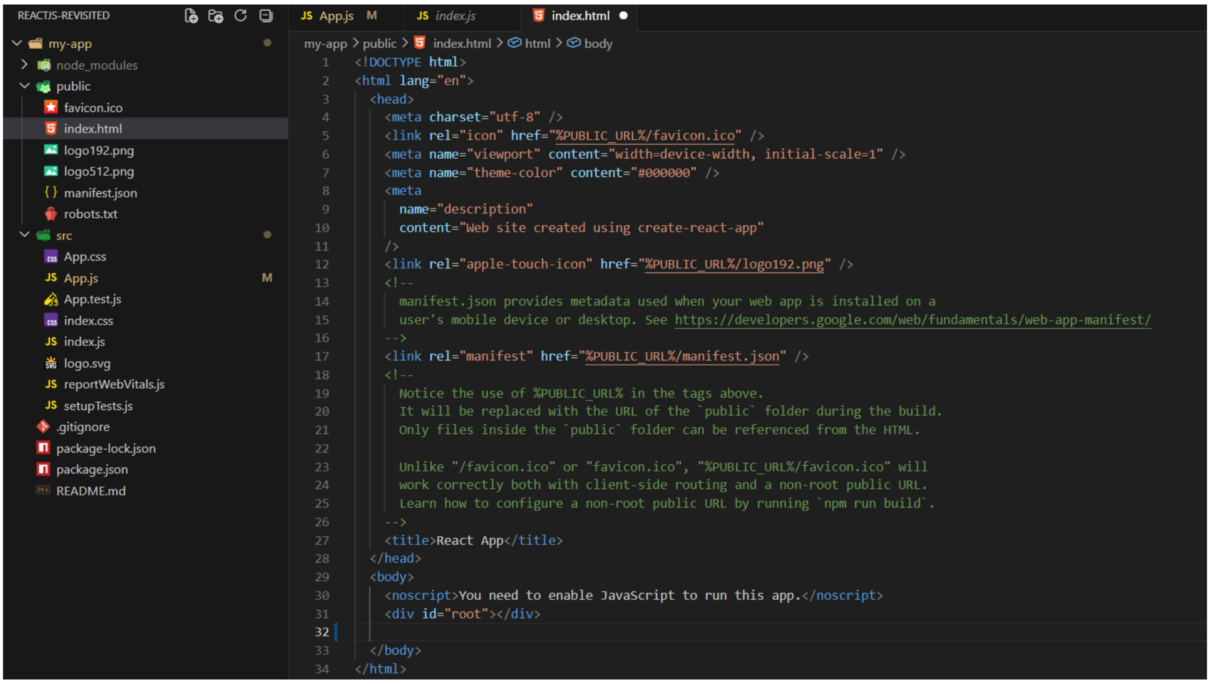
Inside the index.html of the public folder, as shown below:



Why index.html?

Because for the SPA, in react index.html is the only page which will be loading every time.

Before pasting the link:



After pasting the link:

```html
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="utf-8" />
  <link rel="icon" href="%PUBLIC_URL%/favicon.ico" />
  <meta name="viewport" content="width=device-width, initial-scale=1" />
  <meta name="theme-color" content="#000000" />
  <meta name="description" content="Web site created using create-react-app" />
  <link rel="apple-touch-icon" href="%PUBLIC_URL%/logo192.png" />
  <!--
      manifest.json provides metadata used when your web app is installed on a
      user's mobile device or desktop. See https://developers.google.com/web/fundamentals/web-app-manifest/
    -->
  <link rel="manifest" href="%PUBLIC_URL%/manifest.json" />
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/css/bootstrap.min.css" rel="stylesheet"
    integrity="sha384-sRIl4kxILFvY47J16cr9ZwB07vP4J8+LH7qKQnuqkuIAvNWLzeN8tE5YBujZqJLB" crossorigin="anonymous">

  <title>React App</title>
</head>

<body>
  <noscript>You need to enable JavaScript to run this app.</noscript>
  <div id="root"></div>
  <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/js/bootstrap.bundle.min.js"
    integrity="sha384-FKyoEForCGlyvwx9Hj09JcYn3nv7wiPVlz7YYwJrWVcXK/BmnVDxM+D2scQbITxI"
    crossorigin="anonymous"></script>
</body>

</html>
```

Now, let's do something in the code to see the implementation of them practically.

As usual we will work on the App.js:

```javascript
import logo from './logo.svg';
import './App.css';

function App() {
  return (
    <div className="container mt-5">
      <h1>React + Bootstrap</h1>

      <button className="btn btn-primary">
        Click Me
      </button>
    </div>
  );
}

export default App;
```

Output: (don't forget to do 'npm start')

What is the difference between JS function call and React function call:

- JS function → you call it anytime, it runs immediately.
- React component function → React calls it during render to build UI, and may re-call it on state/props changes.

Example:

App.js:

```
function add(a, b) {
  return a + b;
}
console.log(add(2, 3)); // 5
```

Runs immediately when called.

App.jsx:

```
function Hello() {
  return <h1>Hello World</h1>;
}
export default function App() {
  return <Hello />;
}
```

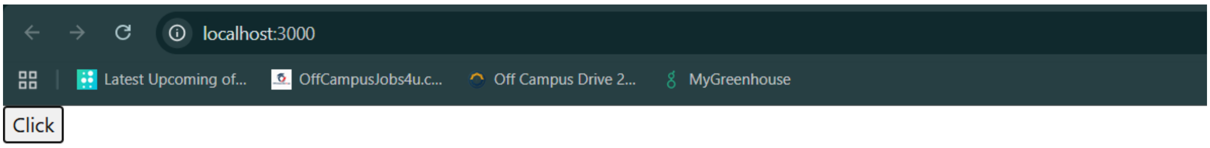
React calls Hello() internally during rendering, not you directly like normal JS logic.

How to pass functions to the button for onClick event?

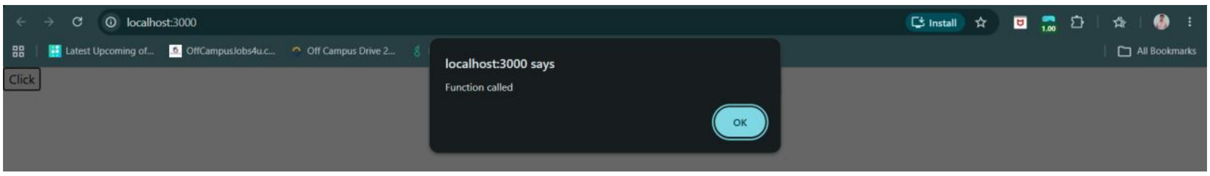
App.js:

```
src > JS App.js > default
function App(){
  function callFun(){
    alert("Function called");
  }
  return(
    <>
    <button onClick={callFun}>Click</button>
    </>
  )
}
export default App;
```

Output:



When clicked on button:



Basically, Always pass a function reference, not a function call.

Wrong way:

src > JS App.js > ...

```
function App(){
  function callFun(){
    alert("Function called");
  }
  return(
    <>
    <button onClick={callFun()}>Click</button>
    </>
  )
}
export default App;
```

Output:

Not worked as expected.